AIML x Astrophysics: Session Overview

A beginner-friendly session where machine learning meets the stars.

What this session is about

Stars look like a messy sky of bright dots, but they follow clear patterns. In this session we first learn how a machine learns patterns from data, then use that idea to make sense of stars. By the end, you will see why an algorithm called **K-Nearest Neighbours (K-NN)** is a natural fit for sorting stars into groups.

No prior ML or physics background is needed. Everything is explained with everyday examples and visuals.

Who it is for

Anyone curious about AI/ML or astronomy. If you can read a graph, you are ready.

What you will take away

- A clear idea of what AI and ML are, and how supervised and unsupervised learning differ
- An intuition for "data space" and why some patterns need smarter methods than a straight line
- How computers measure "closeness" between data points
- How K-NN classifies a new point by looking at its neighbours
- Basic stellar physics: spectra, star types, luminosity and the HR diagram
- How the two halves connect: classifying stars with K-NN

Session flow

Part 1: Machine learning foundations

1. What are AI and ML? A short theory recap of what AI and ML mean, and how they relate.

2. Supervised vs unsupervised learning

- *Supervised:* we already know the answers, so we classify or predict. Examples: voice to text, predicting share prices.
- *Unsupervised:* we look for structure hidden in the data, by clustering or reducing complexity. Example: shopping recommendations on Amazon or Flipkart.

3. What is data space? A visual explanation of how data points sit in space and form patterns. We look at three common shapes: **bands, rings and islands**. A single straight line can fail on bands and rings, while K-NN handles all three by looking at the local neighbourhood.

4. Distance measures How does a computer decide that two points are "close"? We explain it with real-life examples:

- **Euclidean (L2):** straight-line distance, like walking across a playground
- **Manhattan (L1):** moving along a grid, like a chessboard or the streets of Manhattan
- **Chebyshev (p = ∞):** only the largest single step counts

We also visualise each one as a shape: the set of points at distance 1 from the centre is a diamond, a circle and a square respectively.

5. What is K-NN? A new data point takes the class of the neighbours around it. We discuss why this method suits our dataset.

6. Choosing K (optional) A small K can overfit and chase noise, while a large K can underfit and blur real differences. We will cover this only if the group is comfortable with it.

Part 2: The physics of stars

1. What is a star? A quick introduction to the objects we are about to classify.

2. Stellar spectra

- A hot, dense core gives a continuous glow, and the cooler outer atmosphere absorbs specific wavelengths, leaving dark lines.
- Electrons jump between fixed energy levels, so each element leaves a unique line "fingerprint".

3. Star classification (O B A F G K M) Stars are sorted by colour and temperature into classes, each split into subclasses 0 to 9. Our Sun is a G2 star.

4. Luminosity A star's brightness depends on both its size and its temperature. A big, cool star can outshine a small, hot one, as with Betelgeuse versus Sirius B.

5. The HR diagram The Hertzsprung-Russell diagram plots luminosity against temperature.

- Hot stars sit on the left and cool stars on the right (a historical quirk).
- Stars cluster into groups instead of scattering randomly.
- We meet the main sequence and the other families of stars on the diagram.

The big idea that ties it together

The HR diagram is a data space. Stars fall into natural groups, and a new star can be classified by checking which stars sit closest to it. That is exactly what K-NN does.

Before you come

- Bring curiosity; no preparation is required
- Optional: think of one example of something a computer predicts or recommends in your daily life

Date, time and venue: to be shared separately.